THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

CIVL2108: PRINCIPLES OF CIVIL ENGINEERING MANAGEMENT

May 18, 2015 2:30 pm – 5:30 pm (3 hours)

Answer **ANY FOUR** questions

All questions carry equal marks

Use of Calculators

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

1. The government intends to use Kowloon East as a pilot area to explore the feasibility of developing a Smart City. To trial the Smart City concept, a series of construction projects of different types and scales will be rolled out in the region for the years to come.

   a) Explain the mechanism of two-stage selective tendering and discuss the pros and cons of this tendering approach over conventional selective tendering method when applied to the Smart City projects. [15 marks]

   b) Outline the principle of serial tendering and discuss its suitability for developing a series of Smart City related construction projects in the Kowloon East region. [10 marks]

2. The following table shows the duration and logical relationship of the activities involved in a construction project:

Activity ID	*Predecessor*	*Duration (days)*
A	--	10
B	A	30
C	B	30
D	C (SS10)	40
E	D (SS10)	40
F	C	40
G	D	28
H	E	30
I	D	30
J	E	30
K	G, H, I, J	70
L	D, E, F	60
M	K	100
N	L, M	10

   a) Prepare an activity-on-node network diagram based on the information given in the table above and identify the critical path of this project. [10 marks]

   b) Prepare a table showing the early and late start and completion dates of each activity and work out the total and free floats accordingly. [8 marks]

   c) Describe what measures should be taken by the project manager to monitor and control project progress. [7 marks]

3. a) With reference to the roles and responsibilities of civil engineers, explain how civil engineers can serve competently, collaboratively and ethically as master planners, designers, constructors and operators of society's economic and social engine. [17 marks]

b) Discuss with examples what unethical behaviours consultant engineers should be aware of when tenderers are selected from the approved list of contractors. What action(s) should be taken should these unethical behaviours be discovered? [8 marks]

4. a) Calculate the unit rate of an in-situ concrete retaining wall based on the cost data as shown below and your own assumptions (if necessary). The wall is tapered in section with its thickness at the top and bottom being 200mm and 280mm respectively. The concrete is made up of one part of cement, one-and-a-half parts of sand and three parts of aggregate.

Labour

Type	*Wage*
Labour for concrete vibration	$120.00 / hour
All other workers required	$100.00 / hour

Location	*Labour Constants*	
	Man hour	*Gang hour*
Wall 150 – 300mm thick	6.32	1.58

Material

Type	*Material price*
Cement	$800.00 / tonne
Sand	$120.00 / tonne
Aggregate	$100.00 / tonne

Plant

Type	*Hiring Charge*	*Output*
Mixer	$150.00 / hour	5 m^3/hour (~50% utilisation)

Note: According to previous cost estimation records, some estimators have added 15% extra labour time when calculating the unit rates of in-situ concrete walls with tapered sectional shapes while the others have made no additional provision for this type of walls. [20 marks]

b) Discuss with reasons whether it is reasonable to have the unit rate of a 100mm thick in-situ concrete wall cheaper than that of a 200mm thick wall if the concrete mix is identical. In case of any differences in the unit rates due to the wall thickness, explain what is the contributing factor and why? [5 marks]

5. a) A hotel owner would like to evaluate the maximum value of a piece of land for hotel development. Outline and compare different financial feasibility methods and recommend with justifications the most suitable method for calculating the land value to support the hotel owner's auctioning decision. [7 marks]

b) Describe the major components of the chosen financial feasibility method in (a) and highlight the assumptions to be considered when calculating the maximum land price. [13 marks]

b) Discuss the risks that could affect the success of the hotel investment. [5 marks]

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